

```
class ClockGUI implements ActionListener, WindowListener {
    private int total_log = 0;

    //frame
    private JFrame frame = new JFrame("Clock");
    private JFrame log_window = new JFrame("Logs Window");

    //panel
    private JPanel container = new JPanel(new BorderLayout());
    private JPanel save_container = new JPanel(new FlowLayout());
    private JPanel clock_p = new JPanel();
    private JPanel output_container = new JPanel(new BorderLayout());
    private JPanel log_status = new JPanel(new BorderLayout());
    //thread
    private Thread t;

    //components
    private JTextArea log = new JTextArea(10,50);
    private JTextField input = new JTextField(50);
    private JButton clear_log = new JButton("Clear Logs");
    private JButton save = new JButton("Save");
    private JLabel total_text = new JLabel("Total Logs: " + total_log);

    //gui
    public ClockGUI() {
        save.addActionListener(this);
        t = new Thread(clock_p);
        t.start();
        clock_p.setBorder(new EmptyBorder(10,10,10,10));
        save_container.add(input);
        save_container.add(save);
        container.add(clock_p, BorderLayout.CENTER);
        container.add(save_container, BorderLayout.SOUTH);
        frame.add(container);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setResizable(false);
        frame.pack();
        frame.setVisible(true);

        log.setEditable(false);
        clear_log.addActionListener(this);
        log_status.add(total_text, BorderLayout.WEST);
        log_status.add(clear_log, BorderLayout.EAST);
        output_container.add(log, BorderLayout.CENTER);
        output_container.add(log_status, BorderLayout.SOUTH);
        log_window.add(output_container);
        log_window.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        log_window.setResizable(false);
        log_window.pack();
        log_window.setVisible(true);

        log_window.addWindowListener(this);
        frame.addWindowListener(this);
    }
}
```

1

```
@Override
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == save) {
        DateTimeFormatter dateFormatter = DateTimeFormatter.ofPattern("yyyy/MM/dd");
        DateTimeFormatter timeFormatter = DateTimeFormatter.ofPattern("HH:mm:ss");
        LocalDateTime now = LocalDateTime.now();
        String date = now.format(dateFormatter);
        String time = now.format(timeFormatter);

        if (log.getText().isEmpty()) {
            log.setText(date + " " + time + ": " + input.getText());
        } else {
            log.setText(log.getText() + "\n" + date + " " + time + ": " + input.getText());
        }

        input.setText("");
        total_log += 1;
        total_text.setText("Total Logs: " + total_log);
    }

    else if (e.getSource() == clear_log) {
        total_log = 0;
        total_text.setText("Total Logs: " + total_log);
        log.setText("");
    }
}
```

3

กดปุ่ม save & clear logs

```
public class Dataobj implements Serializable {
    private String text = "";
    private int total = 0;
    public Dataobj(String text, int total) {
        this.text = text;
        this.total = total;
    }
    public String getText() {
        return text;
    }
    public int getTotal() {
        return total;
    }
}
```

แยกไฟล์

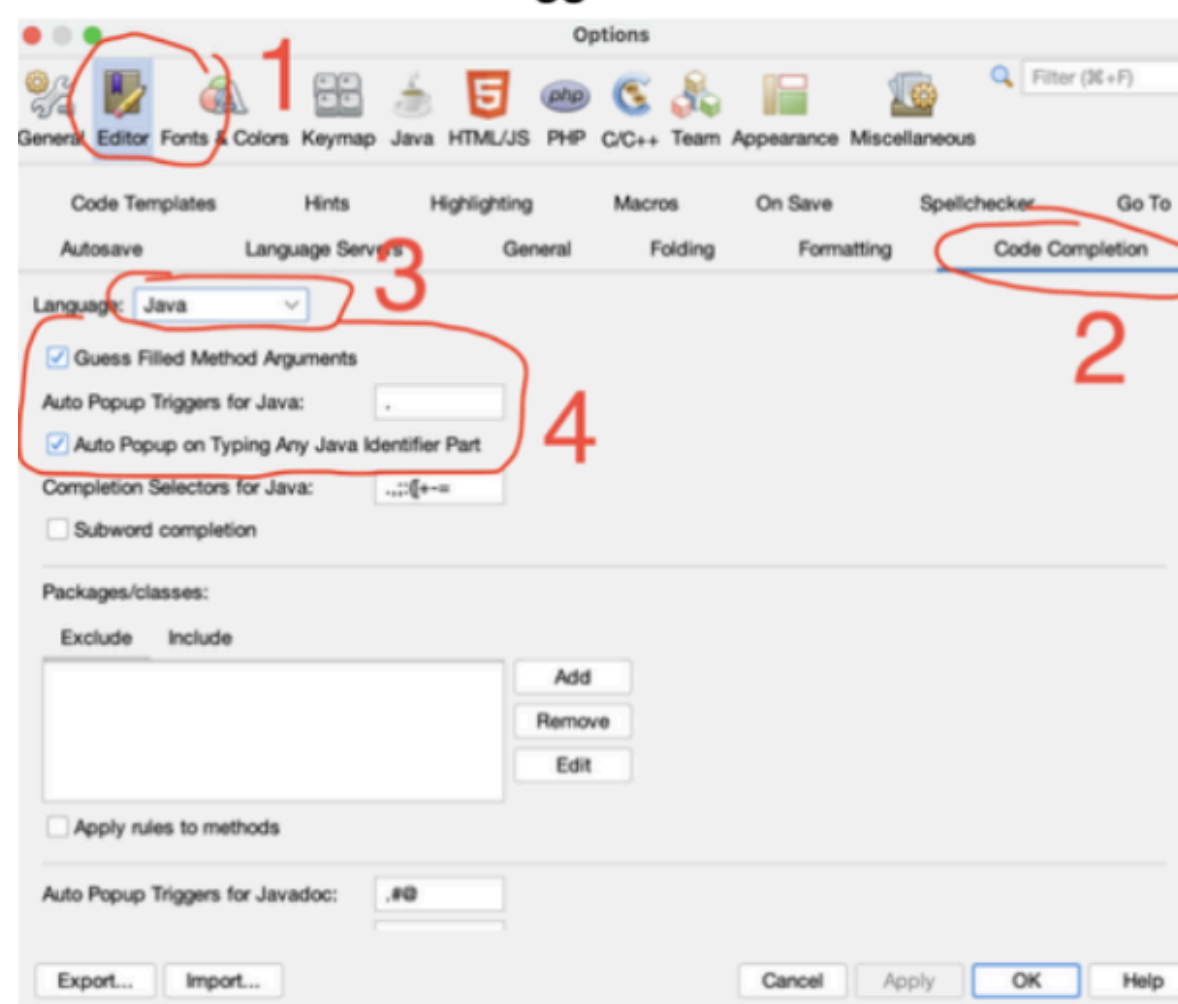
```
@Override
public void windowOpened(WindowEvent e) {
    File f = new File("logs.txt");
    if (!f.exists()) return;
    try (ObjectInputStream in = new ObjectInputStream(new FileInputStream("logs.txt"))) {
        Dataobj data = (Dataobj) in.readObject();
        total_log = data.getTotal();
        log.setText(data.getText());
        total_text.setText("Total Logs: " + total_log);
    } catch (ClassNotFoundException | FileNotFoundException ex) {
        System.getLogger(ClockGUI.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
    } catch (IOException ex) {
        System.getLogger(ClockGUI.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
    }
}

@Override
public void windowClosing(WindowEvent e) {
    Dataobj obj = new Dataobj(log.getText(), total_log);
    try (ObjectOutputStream out = new ObjectOutputStream(new FileOutputStream("logs.txt"))) {
        out.writeObject(obj);
    } catch (IOException ex) {
        ex.printStackTrace();
    }
}
```

5

เขียนไฟล์ออก

TIP: พิมพ์แล้วขึ้น auto suggest ให้เลย



```
go_left.addActionListener(e -> {
    int current = Integer.parseInt(book_order.getText());
    if (current > 1) {
        int targetIndex = current - 2;
        Book b = (Book) data.get(targetIndex);
        tf_name.setText(b.getName());
        tf_price.setText(b.getPrice() + "");
        cb_type.setSelectedItem(b.getType());
        book_order.setText(current - 1 + "");
    }
});

go_right.addActionListener(e -> {
    int current = Integer.parseInt(book_order.getText());
    if (current < data.size()) {
        int targetIndex = current;
        Book b = (Book) data.get(targetIndex);
        tf_name.setText(b.getName());
        tf_price.setText(b.getPrice() + "");
        cb_type.setSelectedItem(b.getType());
        book_order.setText(current + 1 + "");
    }
});

@Override
public void windowOpened(WindowEvent e) {
    File file_check = new File("Book.data");
    if (!file_check.exists()) {
        tf_name.setText("");
        tf_price.setText("");
        cb_type.setSelectedIndex(0);
        book_order.setText("");
        return;
    }
    try (ObjectInputStream inp = new ObjectInputStream(new FileInputStream("Book.data"))) {
        data = (ArrayList) inp.readObject();
        Book in = (Book) data.get(0);
        tf_name.setText(in.getName());
        tf_price.setText(in.getPrice() + "");
        cb_type.setSelectedItem(in.getType());
        book_order.setText("1");
    } catch (FileNotFoundException ex) {
        System.getLogger(BookView.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
    } catch (IOException | ClassNotFoundException ex) {
        System.getLogger(BookView.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
    }
}

public BookAdd(BookView parent) {
    cb_type.setSelectedIndex(0);

    sub_con.add(tf_name);
    sub_con.add(tf_price);
    sub_con.add(tf_price);
    sub_con.add(tf_type);
    sub_con.add(cb_type);
    container.add(sub_con, BorderLayout.CENTER);

    order_con.add(go_left);
    order_con.add(book_order);
    order_con.add(go_right);

    util_con.add(b_add);
    util_con.add(b_update);
    util_con.add(b_delete);
    sub_con2.add(order_con);
    sub_con2.add(util_con);

    container.add(sub_con2, BorderLayout.SOUTH);
    frame.add(container);
    frame.setVisible(true);
    frame.pack();
    frame.addWindowListener(this);
}
```

3

PARENT

```
public class BookView implements WindowListener {
    ArrayList data = new ArrayList();

    private JFrame frame = new JFrame("");
    private JPanel container = new JPanel(new BorderLayout());
    private JPanel sub_con = new JPanel(new GridLayout(3, 2));
    private JPanel sub_con2 = new JPanel(new FlowLayout(2, 1));
    private JPanel order_con = new JPanel(new FlowLayout());
    private JPanel util_con = new JPanel(new FlowLayout());

    private JLabel t_name = new JLabel("Name");
    private JLabel t_price = new JLabel("Price");
    private JLabel t_type = new JLabel("Type");

    private String[] type_list = {"General", "Computer", "Math&Sci", "Photo"};

    JTextField tf_name = new JTextField();
    JTextField tf_price = new JTextField();
    JComboBox cb_type = new JComboBox(type_list);
    JTextField book_order = new JTextField(5);
    private JButton go_left = new JButton("<<");
    private JButton go_right = new JButton(">>");
    private JButton b_add = new JButton("Add");
    private JButton b_update = new JButton("Update");
    private JButton b_delete = new JButton("Delete");

    public BookView() {
        b_add.addActionListener((ActionEvent e) -> {
            BookAdd bookAdd = new BookAdd(BookView.this);
        });

        b_update.addActionListener((ActionEvent e) -> {
            Book selected = (Book) data.get(Integer.parseInt(book_order.getText()) - 1);
            selected.setName(tf_name.getText());
            selected.setPrice(Double.parseDouble(tf_price.getText()));
            selected.setType((String) cb_type.getSelectedItem());
            JOptionPane.showMessageDialog(frame, "Done it.");
        });

        b_delete.addActionListener((ActionEvent e) -> {
            Book selected = (Book) data.remove(Integer.parseInt(book_order.getText()) - 1);
            if (!data.isEmpty()) {
                Book b = (Book) data.get(0);
                tf_name.setText(b.getName());
                tf_price.setText(b.getPrice() + "");
                cb_type.setSelectedItem(b.getType());
                book_order.setText("1");
            } else {
                tf_name.setText("");
                tf_price.setText("");
                cb_type.setSelectedIndex(0);
                book_order.setText("0");
            }
            JOptionPane.showMessageDialog(frame, "Done it.");
        });
    }
}
```

1

2

COMBOBOX

```
public class BookAdd {
    private JFrame frame = new JFrame("");
    private JPanel container = new JPanel(new BorderLayout());
    private JPanel sub_con = new JPanel(new GridLayout(3, 2));
    private JPanel sub_con2 = new JPanel(new FlowLayout());

    private JLabel t_name = new JLabel("Name");
    private JLabel t_price = new JLabel("Price");
    private JLabel t_type = new JLabel("Type");
    private String[] type_list = {"General", "Computer", "Math&Sci"};
    private JTextField tf_name = new JTextField();
    private JTextField tf_price = new JTextField();
    private JComboBox cb_type = new JComboBox(type_list);
    private JButton b_insert = new JButton("Insert");

    public BookAdd(BookView parent) {
        cb_type.setSelectedIndex(0);

        sub_con.add(tf_name);
        sub_con.add(tf_price);
        sub_con.add(tf_price);
        sub_con.add(tf_type);
        sub_con.add(cb_type);
        container.add(sub_con, BorderLayout.CENTER);

        b_insert.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                Book b = new Book(tf_name.getText(), Double.parseDouble(tf_price.getText()), (String) cb_type.getSelectedItem());
                parent.data.add(b);
                parent.showDialog(frame, "Done it.");
                parent.tf_name.setText(b.getName());
                parent.tf_price.setText(b.getPrice() + "");
                parent.cb_type.setSelectedItem(b.getType());
                parent.book_order.setText(parent.data.size() + "");
            }
        });

        frame.dispose();
    }

    sub_con2.add(b_insert);
    container.add(sub_con2, BorderLayout.SOUTH);
    frame.add(container);

    frame.setVisible(true);
    frame.pack();
}
```

```
public class ClockPanel extends JPanel implements Runnable {
    private JLabel datetext = new JLabel();
    private JLabel timetext = new JLabel();

    public ClockPanel() {
        this.setLayout(new GridLayout(2, 1));
        datetext.setHorizontalAlignment(JLabel.CENTER);
        timetext.setHorizontalAlignment(JLabel.CENTER);
        timetext.setFont(new Font("Serif", Font.PLAIN, 50));
        this.add(datetext);
        this.add(timetext);
    }

    @Override
    public void run() {
        while (true) {
            DateTimeFormatter dateFormatter = DateTimeFormatter.ofPattern("yyyy/MM/dd");
            DateTimeFormatter timeFormatter = DateTimeFormatter.ofPattern("HH:mm:ss");
            LocalDateTime now = LocalDateTime.now();
            String date = now.format(dateFormatter);
            String time = now.format(timeFormatter);

            datetext.setText(date);
            timetext.setText(time);

            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}
```

แยกไฟล์

Thread

```
public class Book implements Serializable {
    private String name;
    private double price;
    private String type;

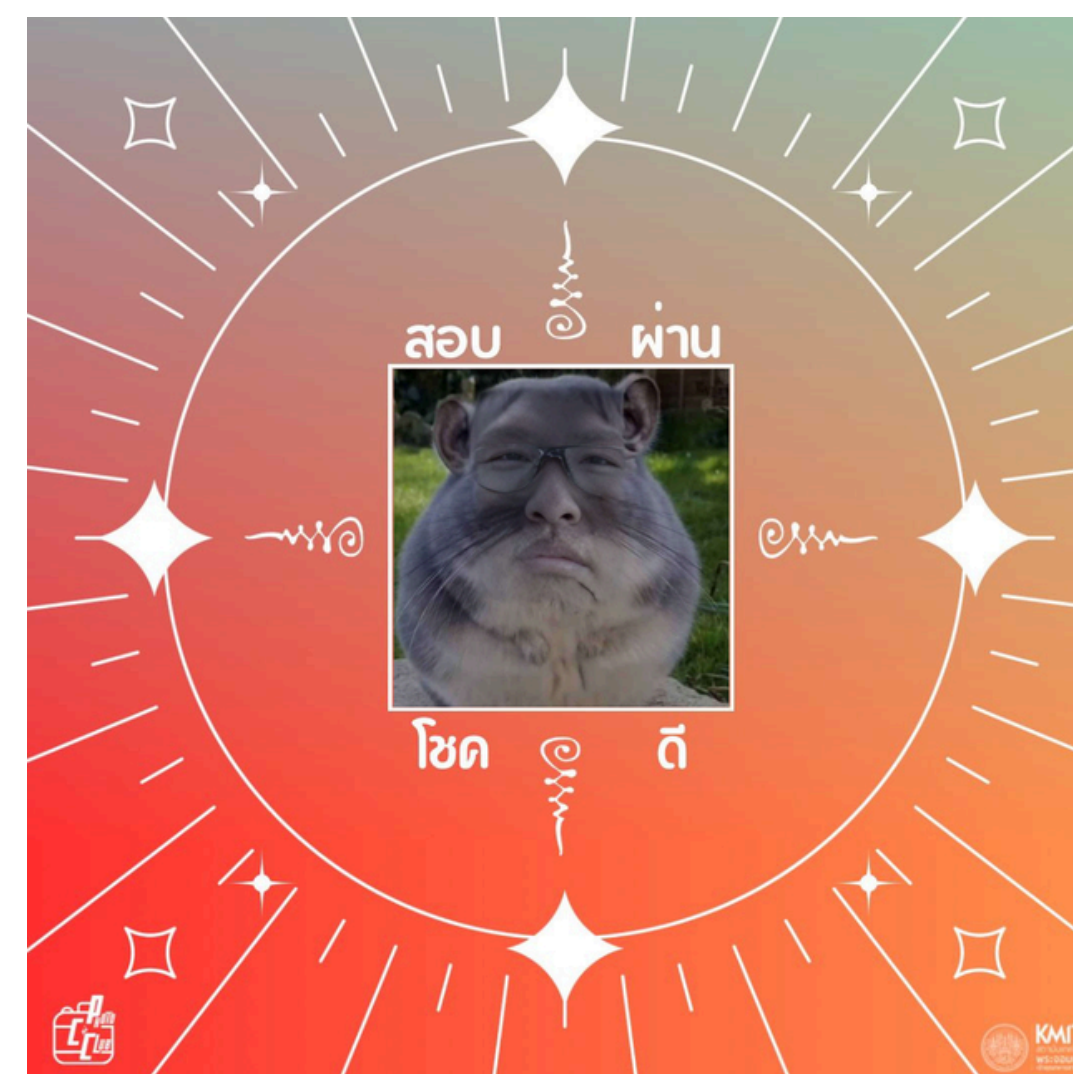
    public Book(String name, double price, String type) {
        this.name = name;
        this.price = price;
        this.type = type;
    }

    public Book() {
        this.name = "";
        this.price = 0.00;
        this.type = "";
    }
}
```

แยกไฟล์

IMPLEMENT SERIALIZABLE ก่อน SAVE OBJECT

```
@Override
public void windowClosing(WindowEvent e) {
    try (ObjectOutputStream out = new ObjectOutputStream(new FileOutputStream("Book.data"))) {
        if (!data.isEmpty()) {
            out.writeObject(data);
        }
    } catch (FileNotFoundException ex) {
        System.getLogger(BookView.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
    } catch (IOException ex) {
        System.getLogger(BookView.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
    }
}
```



6


```
private int overalltime;
private boolean paused = false;

public void pauseClock() {
    paused = true;
}

public void resumeClock() {
    paused = false;
}

@Override

public void run() {
    while (true){
        if(!paused){
            int hour = overalltime / 3600;
            int min = (overalltime % 3600) / 60;
            int sec = overalltime%60;
            String timeText = String.format("%02d : %02d : %02d", hour, min, sec);
            this.setText(timeText);
            this.setFont(new Font("Serif", Font.BOLD, 18));
            overalltime++;

        }

        try {
            Thread.sleep(1000);
        } catch (InterruptedException ex) {
            System.getLogger(MyClock.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
        }
    }
}
```

```
public class MyClock extends JLabel implements Runnable{
```

```
@Override
public void run() {

    while (true){
        Calendar d = Calendar.getInstance();
        int sec = d.get(Calendar.SECOND);
        int min = d.get(Calendar.MINUTE);
        int hour = d.get(Calendar.HOUR_OF_DAY);
        String timeText = String.format("%02d : %02d : %02d", hour, min, sec);
        this.setText(timeText);
        this.setFont(new Font("Serif", Font.BOLD, 18));

        try {
            Thread.sleep(1000);
        } catch (InterruptedException ex) {
            System.getLogger(MyClock.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
        }
    }
}
```

```
private int overalltime;
@Override
public void run() {
    while (true){

        int hour = overalltime / 3600;
        int min = (overalltime % 3600) / 60;
        int sec = overalltime%60;
        String timeText = String.format("%02d : %02d : %02d", hour, min, sec);
        this.setText(timeText);
        this.setFont(new Font("Serif", Font.BOLD, 18));

        try {
            Thread.sleep(1000);
            overalltime++;
        } catch (InterruptedException ex) {
            System.getLogger(MyClock.class.getName()).log(System.Logger.Level.ERROR, (String) null, ex);
        }
    }
}
```

Components

```
- JMenu JMenuBar JMenuItem
JMenuBar mb = new JMenuBar();
menu = new JMenu("Menu");
submenu = new JMenu("Sub Menu");
i1 = new JMenuItem("Item 1");
i2 = new JMenuItem("Item 2");
i3 = new JMenuItem("Item 3");
i4 = new JMenuItem("Item 4");
i5 = new JMenuItem("Item 5");
menu.add(i1);
menu.add(i2);
menu.add(i3);
submenu.add(i4);
submenu.add(i5);
menu.add(submenu);
mb.add(menu);
frame.setJMenuBar(mb);

menu.addSeparator(); --> เว้นบรรทัด
```

- JComboBox

```
String list[] = {"C","C+","C#","Java","PHP"};
JComboBox cb = new JComboBox(list);
cb.getItemAt(cb.getSelectedIndex()) --> เอาไอเทมจาก index
cb.addItem("FewFewFew"); --> เพิ่มไอเทม
cb.setSelectedItem("C"); --> เลือกไอเทมนั้นๆ
```

- JRadioButton

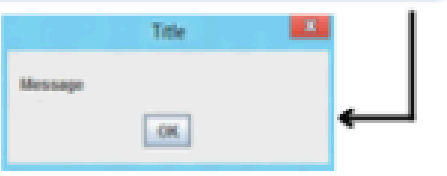
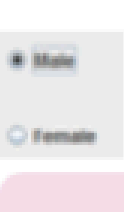
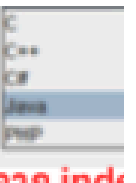
```
JRadioButton r1 = new JRadioButton("Male");
JRadioButton r2 = new JRadioButton("Female");
ButtonGroup bg = new ButtonGroup();
bg.add(r1); bg.add(r2); --> จัดกลุ่มปุ่ม ให้เลือกได้แค่ปุ่มเดียว
```

- JOptionPane

```
JOptionPane.showMessageDialog(
    null, "Message", "Title", JOptionPane.PLAIN_MESSAGE);
```

- JCheckBox

```
ck = new JCheckBox("Check me!");
ck.setSelected(true);
boolean selected = ck.isSelected();
```



```
private void saveFile() {
    try (PrintWriter pw = new PrintWriter(new File(pathname: "ChatDemo.dat")))
        pw.print(= ta.getText());
    } catch (IOException e) {
        e.printStackTrace();
    }
}

private void loadFile() {
    File file = new File(pathname: "ChatDemo.dat");
    if (file.exists()) {
        try (BufferedReader br = new BufferedReader(new FileReader(file))) {
            String line;
            while ((line = br.readLine()) != null) {
                ta.append(line + "\n");
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    desktop = new JDesktopPane();
    mb = new JMenuBar();

    // (Internal Frames)
    frame1 = new JInternalFrame(title: "Appli");
    frame1.setSize(width: 200, height: 150);
    frame1.setLocation(x: 20, y: 250);
    frame1.setVisible(aFlag: true);

    frame2 = new JInternalFrame(title: "Appli");
    frame2.setSize(width: 250, height: 180);
    frame2.setLocation(x: 250, y: 180);
    frame2.setVisible(aFlag: true);

    frame3 = new JInternalFrame(title: "Appli");
    frame3.setSize(width: 300, height: 220);
    frame3.setLocation(x: 550, y: 300);
    frame3.setVisible(aFlag: true);

    desktop.add(comp: frame1);
    desktop.add(comp: frame2);
    desktop.add(comp: frame3);
    desktop.setBackground(bg: Color.BLACK);

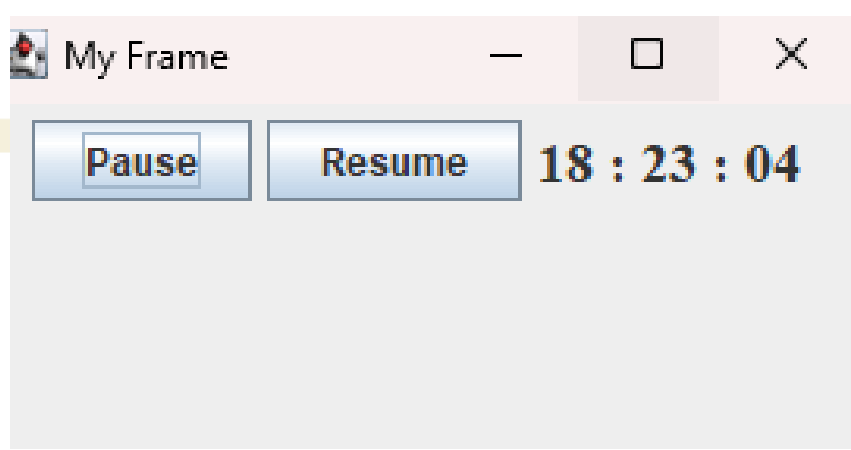
    JButton insBtn = new JButton(text: "Insert");
    insBtn.addActionListener(ev -> {
        System.out.println(x: "Click Insert & Store at Last Index is list");
        BookAdd.dispose();
        JOptionPane.showMessageDialog(parentComponent:null, message:"Done it.");
    });

    public void readFile(File fi) {
        try (BufferedReader br = new BufferedReader(new FileRea
            String line;
            while ((line = br.readLine()) != null) {
                ta.append(line + "\n");
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    public void saveFile(File fi) {
        try (PrintWriter pw = new PrintWriter(new FileWriter(fil
            pw.print(= ta.getText());
        } catch (IOException e){
            e.printStackTrace();
        }
    }

    (Title, ที่เหลือ True หมดเลย)

    desktop.add(comp: frame1);
    desktop.add(comp: frame2);
    desktop.add(comp: frame3);
    desktop.setBackground(bg: Color.BLACK);
```



```
JMenu.addSeparator();
```

```
JButton btn = new JButton("Submit");
btn.addActionListener(e -> {
    if(male.isSelected()){
        System.out.println("Male");
    }
});
```

```
public class MyFrame{
    private JFrame frame;
    private JPanel container;
    private MyClock clock;
    private JButton pause, resume;
    private Thread t;
    public MyFrame () {
        frame = new JFrame("My Frame");
        container = new JPanel();
        clock = new MyClock();
        t = new Thread(clock);

        t.start();
        pause = new JButton("Pause");
        resume = new JButton("Resume");
        pause.addActionListener((ActionEvent e) -> {
            clock.pauseClock();
        });
        resume.addActionListener((ActionEvent e) -> {
            clock.resumeClock();
        });
    }

    container.add(pause);
    container.add(resume);
    container.add(clock);
    frame.add(container);
    frame.setSize(300,150);
    frame.setVisible(true);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}

public static void main(String[] args) {
    new MyFrame();
}
```

การใช้ File writer

```
public void saveFile(String content) {
    try (FileWriter fw = new FileWriter(currentSelectFile)) {
        for (int i = 0; i < content.length(); i++) {
            fw.write(content.charAt(i));
        }
    } catch (IOException ex) {
        ex.printStackTrace();
    }
}

Collection Method
- boolean add (Object element)
- boolean remove (Object element)
- int size()
- boolean isEmpty()
- boolean contains (Object element)
- Iterator iterator()
```

การใช้ File reader

```
public String openFile(File file) {
    this.currentSelectFile = file;
    String text = "";
    try (FileReader fileReader = new FileReader(currentSelectFile)) {
        int ch;
        while ((ch = fileReader.read()) != -1) {
            text += (char) ch;
        }
    } catch (IOException ex) {
        ex.printStackTrace();
    }
    return text;
}
```



หลักการเขียน synchronized

```
private int time;
private boolean paused = false;
@Override
public void run() {
    while (true) {
        synchronized(this) {
            while (paused) {
                try {
                    wait();
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
            int sec = time % 60;
            int min = (time / 60) % 60;
            int hour = (time / 3600) % 24;
            setText(String.format("%02d:%02d:%02d",
                hour, min, sec));
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
            time++;
        }
        public synchronized void pauseThread() {
            paused = true;
        }
        public synchronized void resumeThread() {
            paused = false;
            notify();
        }
    }
}
```

```
JFrame frame = new JFrame();
TimerCanStopClock clock = new TimerCanStopClock();
Thread thread = new Thread(clock);
thread.start();
frame.setSize(450, 150);
frame.addMouseListener(new MouseListener() {
    @Override
    public void mouseClicked(MouseEvent e) {
        isPaused = !isPaused;
        if (isPaused) {
            clock.pauseThread();
        } else {
            clock.resumeThread();
        }
    }
})
```

```
ตอนเรียกใช้ synchronized
boolean isPause = false;
isPaused = !isPaused;
if (isPaused) {
    clock.pauseThread();
} else {
    clock.resumeThread();
}
```



"หายไปสัก ๓๐ วัน กลับมาศึกษา"
อ.ธราวิชญ์ อธิติบุญเรือน

เมธอดที่ต้องลงทะเบียนสำหรับการจัดการกับเหตุการณ์

อินเตอร์เฟส	เมธอดที่ต้อง implements
KeyListener	keyPressed (KeyEvent) keyReleased (KeyEvent) keyTyped (KeyEvent)
FocusListener	focusGained (FocusEvent) focusLost (FocusEvent)
AdjustmentListener	adjustmentValueChanged (AdjustmentEvent)
ComponentListener	componentMoved (ComponentEvent) componentHidden (ComponentEvent) componentResized (ComponentEvent) componentShown (ComponentEvent)

เหตุการณ์	คำอธิบาย
KeyEvent	จะถูกสร้างขึ้นในกรณีที่เกิดเหตุการณ์กดคีย์บอร์ดขึ้น
FocusEvent	จะถูกสร้างขึ้นในกรณีที่เกิดเหตุการณ์ผู้ใช้เลือกอุปกรณ์อินพุต (เมาส์) มาใช้กับวัตถุของ GUI
ComponentEvent	จะถูกสร้างขึ้นในกรณีที่เกิดเหตุการณ์วัตถุของ GUI มีการเปลี่ยนแปลง เช่น จัดดี, คำนำหน้า, ขนาด เป็นต้น
ContainerEvent	จะถูกสร้างขึ้นในกรณีที่เกิดเหตุการณ์เพิ่มลด component ลงบน Container
AdjustmentEvent	จะถูกสร้างขึ้นในกรณีที่เกิดเหตุการณ์ปรับระดับตำแหน่งของวัตถุในกลาง ScrollBar หรือ ScrollPane
TextEvent	จะถูกสร้างขึ้นในกรณีที่เกิดเหตุการณ์เปลี่ยนแปลงข้อความในวัตถุของ class TextArea

Event Handling

- Action Event
- คลิกเมาส์ บน JButton
 - กด enter ใน JTextField
 - เลือกเมนูใน JMenuItem
 - กด Double Click ใน JList

```
public void actionPerformed(ActionEvent e){
    statements;
}
```

- Mouse Event
- Pressed >> คลิกเมาส์ค้าง
 - Released >> ปล่อยเมาส์
 - Dragged >> ลากเมาส์
 - Moved >> ขยับเมาส์
 - Clicked >> คลิกเมาส์
 - entered >> เลื่อนเมาส์ไปบน component
 - Exited >> เลื่อนเมาส์ออกจาก component

Onlearn : <https://onlearn.it.kmitl.ac.th/>

Jlearn : <https://jlearn.it.kmitl.ac.th/>

Onlearn : .zip,/.rar/.7zip

Jlearn : .java

ArrayList<String> names = new

ArrayList<>();

names.add("Alice");

names.add("Bob");

names.add("Charlie");

.dispose() ปิดหน้าต่างนั้น
System.exit(0) ปิดโปรแกรม

Delete Data

```
if (source.equals(deleteButton) && Data.data != null) {
    JOptionPane.showMessageDialog(null, "Done");
    Data.data.remove(now);
    if (Data.data.size() == 0) {
        nameTextField.setText("");
        priceTextField.setText("");
        TypeComboBox.setSelectedItem("General");
        pageTextField.setText("0");
    } else if (now == 0) {
        Book bookData = Data.data.get(now);
        nameTextField.setText(bookData.getName());
        priceTextField.setText(String.valueOf(bookData.getPrice()));
        TypeComboBox.setSelectedItem(bookData.getType());
        pageTextField.setText(String.valueOf(bookData.getPage()));
    } else {
        Book bookData = Data.data.get(now - 1);
        nameTextField.setText(bookData.getName());
        priceTextField.setText(String.valueOf(bookData.getPrice()));
        TypeComboBox.setSelectedItem(bookData.getType());
        pageTextField.setText(String.valueOf(bookData.getPage()));
        now -= 1;
    }
}
```

เมธอดที่ต้องลงทะเบียนสำหรับการจัดการกับเหตุการณ์

อินเตอร์เฟส	เมธอดที่ต้อง implements
ActionListener	actionPerformed (ActionEvent)
ItemListener	itemStateChanged (ItemEvent)
MouseListener	mouseClicked (MouseEvent) mouseEntered (MouseEvent) mouseExited (MouseEvent) mousePressed (MouseEvent) mouseReleased (MouseEvent)
MouseMotionListener	mouseMoved (MouseEvent) mouseDragged (MouseEvent)

เมธอดที่ต้องลงทะเบียนสำหรับการจัดการกับเหตุการณ์

อินเตอร์เฟส	เมธอดที่ต้อง implements
WindowListener	windowOpened (WindowEvent) windowClosed (WindowEvent) windowClosing (WindowEvent) windowIconified (WindowEvent) windowDeiconified (WindowEvent) windowActivated (WindowEvent) windowDeactivated (WindowEvent)
ContainerListener	componentAdded (ContainerEvent) componentRemoved (ContainerEvent)
TextListener	textValueChanged (TextEvent)